

Current OS, Processes, Linux & Utilities

Lesson 60 — Processes, Linux and Windows Utilities

JKSSB Computer Awareness

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Deadlock and Its Four Necessary Conditions

- ▶ A **deadlock** is a state in which a set of processes each wait for a resource held by another, so none can proceed.
- ▶ Four conditions must hold together for a deadlock to occur.

Condition	Meaning
Mutual exclusion	A resource is held by only one process at a time.
Hold and wait	A process holds one resource while waiting for others.
No preemption	A resource cannot be forcibly taken away.
Circular wait	Each process waits for a resource held by the next.

Deadlock Recovery by Resource Preemption

- ▶ Once a deadlock is detected, the system must **recover**.
- ▶ One recovery method is **resource preemption**: take a resource away from one process and give it to another so the cycle breaks.
- ▶ Preemption of a resource is thus a **recovery technique**, not one of the four necessary conditions.

Trap

Preemption is **not** one of the four necessary conditions for deadlock. The condition is **no preemption**; preemption itself is used to recover. (PYQ-10)

Priority Inversion

- ▶ **Priority inversion** happens when a **high-priority** task is blocked because a **low-priority** task holds a resource it needs.
- ▶ While the low-priority task holds the resource, a **medium-priority** task can preempt it and run.
- ▶ The high-priority task therefore waits behind a lower-priority one — the priority order is **inverted**.

One line

A high-priority task waits for a low-priority task, which is itself pushed aside by a medium-priority task.

Priority Inheritance: The Fix

- ▶ **Priority inheritance**: while the low-priority task holds the resource, it **temporarily inherits** the high priority of the task waiting for it.
- ▶ It then runs at the inherited priority, releases the resource quickly, and drops back to its own priority.
- ▶ **Round-robin** scheduling by itself does **not** prevent priority inversion. (PYQ-11)

Keep the pair straight

Inversion = the problem; inheritance = the cure.

Starvation and Aging

- ▶ **Starvation** (indefinite blocking): a process is ready to run but is **repeatedly passed over**, so it may wait forever.
- ▶ It commonly affects **low-priority** processes when higher-priority work keeps arriving.
- ▶ **Aging** is the standard cure: gradually **raise the priority** of a waiting process so it eventually runs.

Starvation vs deadlock

A starved process can still run in principle; a deadlocked process **cannot** proceed at all. Starvation is a scheduling problem, deadlock is a resource-cycle problem.

Preemptive Priority Scheduling

- ▶ In **priority scheduling**, the ready process with the **highest priority** is chosen to run.
- ▶ It is **preemptive** when a running process can be **interrupted** the moment a higher-priority process becomes ready.
- ▶ It is **non-preemptive** when the running process keeps the CPU until it finishes or blocks.

Watch out for

Strict preemptive priority scheduling tends to starve low-priority processes unless **aging** is added.

RTOS: Real-Time Operating System

- ▶ **RTOS** stands for **Real-Time Operating System**.
- ▶ It is an operating system designed to respond within a **bounded time**, so that **deadlines** are met.
- ▶ **Hard real-time**: a missed deadline is a failure (a serious or catastrophic result).
- ▶ **Soft real-time**: a missed deadline only degrades quality; the result is still useful.

Trap

An RTOS is defined by **meeting time deadlines**, **not** by being fast in general. A very fast system that misses deadlines is not real-time.

The Linux Kernel

- ▶ The **kernel** is the **core of the operating system**.
- ▶ It manages **processes** (scheduling), **memory**, **devices** (through drivers), and **file systems**.
- ▶ It runs in **kernel space**; ordinary programs run in **user space** and request services through system calls.
- ▶ The Linux kernel is **open-source** (its source can be viewed, changed, and shared).

Contrast

The **shell** is the command interpreter a user talks to; the **kernel** is the core that actually controls the hardware.

/root vs /home

Path	Meaning
/root	The home directory of the superuser (the root user).
/home	The parent directory that normally holds the home directories of ordinary users (for example /home/alice).
/	The root of the whole file system (a single slash), different again from /root.

Trap

/root is the **superuser's home**, not the top of the file system. The top of the file system is / (single slash).

Windows and Linux File Systems

- ▶ A **file system** organises how files and folders are stored and retrieved on a disk.
- ▶ **NTFS** = New Technology File System — the **default for modern Windows**.
- ▶ **FAT32** = an older, widely compatible file system used for small drives and memory cards.
- ▶ **ext4** = the **native and common default file system for Linux**.

Remember

NTFS → Windows; ext4 → Linux; FAT32 → broad compatibility.

Windows Store

- ▶ The **Windows Store** (now the **Microsoft Store**) is the built-in **online marketplace** in Windows.
- ▶ It is used to **search, download, and install** apps for Windows, and to keep them updated.
- ▶ Apps installed this way come from a **trusted catalogue** rather than from any website.

Trap

The Store is a **source of applications**, **not** a disk-cleanup or defragmentation tool. Do not confuse it with the maintenance utilities.

WordPad vs Notepad

Point	Notepad	WordPad
Text	Plain text , no formatting	Basic formatting (bold, italic, fonts, colour)
Typical save	.txt	.rtf (rich text)
Extra	Smallest, fastest text editor	Can insert pictures and objects

Notepad = plain text; WordPad = light rich text — neither is a full word processor like MS Word.

Disk Cleanup

- ▶ **Disk Cleanup** is a built-in Windows utility that **freed disk space**.
- ▶ It removes **unnecessary files**: temporary files, thumbnails, downloaded program files, and the **Recycle Bin** contents.
- ▶ You choose which categories to delete; the files are not part of your live documents.

Use

When a drive is running low on space, Disk Cleanup deletes safe, non-essential files to recover capacity.

Defragmenter and the SSD Caution

- ▶ A **hard disk drive (HDD)** stores a file in **fragments** spread over the disk.
- ▶ The **Disk Defragmenter** rearranges those fragments so each file is **contiguous**, which speeds up reading.
- ▶ **Caution:** do **not** defragment a **solid-state drive (SSD)**.

Why not an SSD?

An SSD has **no moving parts**, so defragmenting gives no speed benefit and only adds **needless writes**, which shorten its life. Windows uses **TRIM** for SSDs instead.
(Lesson scope)

Imaging Fax: Windows Fax and Scan

- ▶ **Windows Fax and Scan** is the built-in imaging and fax utility in Windows.
- ▶ It can **scan** a document (with a scanner) and **send or receive faxes** through a fax modem or fax service.
- ▶ Scanned items can be saved, viewed, and faxed as **images**.

Remember

Imaging Fax in Windows = the **Windows Fax and Scan** app.

xcopy /s

- ▶ **xcopy** is a Windows **command-line** command that copies files and directory trees.
- ▶ The switch **/s** copies the chosen directory **including its subdirectories**, but skips empty ones.
- ▶ Adding **/e** also copies **empty** subdirectories.

Exact term

Remember the command as written: **xcopy /s**. The **/s** means “with subdirectories”.

Lesson 60: Recall

- ▶ Deadlock needs mutual exclusion, hold and wait, no preemption, and circular wait together.
- ▶ **Resource preemption** is a deadlock **recovery** method — not one of the four conditions.
- ▶ **Priority inversion** = a high-priority task waits behind a low-priority holder; **priority inheritance** cures it.
- ▶ Round-robin by itself does **not** prevent priority inversion.
- ▶ **Starvation** = indefinite waiting; **aging** prevents it.
- ▶ Preemptive priority scheduling interrupts a running process for a higher-priority one.
- ▶ **RTOS** = Real-Time Operating System; defined by meeting deadlines (hard vs soft).
- ▶ The Linux **kernel** is the core; `/root` is the superuser's home, `/home` holds users' homes.
- ▶ NTFS = Windows; ext4 = Linux; FAT32 = compatibility.
- ▶ Notepad = plain text; WordPad = basic rich text; Disk Cleanup frees space; do not defragment an SSD; `xcopy /s` copies subdirectories.